Football Match Predictor

Installation & Run Guide

Step-by-step instructions to install, set up, train and run the app on Windows, macOS or Linux.

Item	Details
Repository	https://github.com/shoaibghulam/football-match-predictor
Backend	Python 3.12, FastAPI, SQLite, CatBoost (API on port 8000)
Frontend	React 19, Vite, TypeScript (web app on port 5173)
Time needed	About 45 minutes the first time (most of it is model training)
License	MIT

1. What you are installing

The app shows football matches for any date in your local time zone, with live scores and calibrated **win / draw / loss probabilities**, expected goals, Over 2.5, both-teams-to-score and the most likely scorelines. Predictions come from the project's own models, trained on more than 250,000 club matches from 38 domestic leagues plus the UEFA Champions League, Europa League and Conference League.

It has two parts that run side by side:

Part	Folder	What it does	Address
Backend API	backend/	Data ingest, database, models, REST API, scheduled jobs	http://127.0.0.1:8000
Frontend	frontend/	The website you open in the browser	http://localhost:5173

2. Prerequisites

Install these tools first. Default installer options are fine.

Tool	Version	Download
Git	any	https://git-scm.com
Python	3.12	https://www.python.org/downloads/ (Windows: tick "Add python.exe to PATH")
uv (Python package manager)	0.5 or newer	Run: pip install uv (or see https://docs.astral.sh/uv/)
Node.js (includes npm)	20 or newer	https://nodejs.org (LTS version)
Docker Desktop (optional)	any	https://www.docker.com - only for the Docker method

Open a terminal (Windows: PowerShell or Git Bash; macOS/Linux: Terminal) and check that everything is installed:

```
git --version
python --version
uv --version
node --version
npm --version
```

Each command should print a version number. If one says "not recognized" or "command not found", reinstall that tool and open a new terminal.

3. Installation - first time only

Step 1 - Download the code

```
git clone https://github.com/shoaibghulam/football-match-predictor.git
cd football-match-predictor
```

Step 2 - Add the training data

The historical data is too large for the repository, so you download it once.

1. Sign in to **kaggle.com** (free account) and search the datasets for **"Club Football Match Data (2000 - 2025)"**. Download it.
2. Unzip it and copy **Matches.csv** and **EloRatings.csv** into the **data/raw/** folder:

```
football-match-predictor/
  data/
    raw/
      Matches.csv        (about 45 MB, ~239,000 matches)
      EloRatings.csv     (about 11 MB)
```

Create the **data/raw** folders yourself if they do not exist.

Step 3 - Get a free football-data.org API key

The key unlocks the Champions League schedule, live scores and results. It is free.

1. Register at **https://www.football-data.org/client/register** - the key arrives by email.
2. Create your settings file from the template:

```
cd backend
cp .env.example .env              # macOS / Linux / Git Bash
Copy-Item .env.example .env       # Windows PowerShell
```

3. Open **backend/.env** in a text editor and paste your key after the equals sign:

```
FOOTBALL_DATA_ORG_KEY=paste_your_key_here
```

Keep the key private. backend/.env is git-ignored so it is never uploaded. All other settings in the file are optional and already have sensible defaults.

Step 4 - Install the backend and build the models

Run these commands **inside the backend/ folder**, one at a time. Wait for each to finish.

#	Command	What it does	Time
1	uv sync	Installs Python packages into backend/.venv	1-3 min
2	uv run python -m matchpredict.db.seed	Loads the 238k historical matches into the database	~30 s
3	uv run python -m matchpredict.ingest.europe seed	Downloads 15 seasons of UEFA results	~1 min
4	uv run python -m matchpredict.predictor.train	Trains the domestic model and runs the backtest	~20 min
5	uv run python -m matchpredict.predictor.europe	Trains the European (UEFA) model	few min
6	uv run python -m matchpredict.ingest.run	Fetches the latest results and fixtures	1-2 min
7	uv run python -m matchpredict.ingest.europe refresh	Fetches current UEFA fixtures and scores	< 1 min

Copy-paste version:

```
uv sync
uv run python -m matchpredict.db.seed
uv run python -m matchpredict.ingest.europe seed
uv run python -m matchpredict.predictor.train
uv run python -m matchpredict.predictor.europe
uv run python -m matchpredict.ingest.run
uv run python -m matchpredict.ingest.europe refresh
```

Trained models are saved in **backend/artifacts/** (model.joblib, europe.joblib) with their accuracy reports (metrics.json, europe_metrics.json). Tip: for a quick trial run on less data use **uv run python -m matchpredict.predictor.train --quick**.

Step 5 - Install the frontend

In a terminal, from the project root:

```
cd frontend
npm install
```

4. Running the app

You need **two terminals** open at the same time - one for the backend, one for the frontend.

Terminal 1 - backend API

```
cd football-match-predictor/backend
uv run matchpredict
```

Wait for the line *Uvicorn running on http://127.0.0.1:8000*. Check it works by opening **http://127.0.0.1:8000/api/v1/health** - you should see *"model_loaded": true*. Interactive API documentation is at **http://127.0.0.1:8000/api/v1/docs**.

Terminal 2 - website

```
cd football-match-predictor/frontend
npm run dev
```

Open **http://localhost:5173** in your browser. Use **localhost**, not 127.0.0.1.

Every day after the first install you only need these two commands: **uv run matchpredict** (in backend/) and **npm run dev** (in frontend/). While the backend is running it updates itself: results and fixtures daily at 07:00 UTC, live UEFA scores every 2 minutes during matches, and a weekly model retrain.

Stopping the app

Press **Ctrl + C** in each terminal. On Windows, if a port stays busy, close the terminal window.

Demo mode (no backend, no data)

To preview the website with realistic sample data, without installing the backend:

```
cd frontend
npm install
npm run dev:mock
```

5. Using the app

Page	How to use it
Matches (home)	Shows today's matches in your time zone. Change the day with the date picker, the arrows or the Today / Tomorrow / Next 7 days buttons. Filter by league or search a team. Live and featured matches appear at the top.
Match card	Bar with home / draw / away chances, expected goals, Over 2.5, both teams to score and top scorelines. Click Details for the full view.
Teams	Pick a team to see its upcoming fixtures with its win chance, recent form and ratings.
Predict	Choose any home and away team for an instant prediction.
About	How the model works and how accurate it is.

Accuracy in a 4-season backtest: about 49.5% of results predicted correctly from ratings alone and 50.5% when bookmaker odds are available - level with the bookmaker. Football is uncertain; the app is for information and entertainment, not betting advice.

6. Alternative: run with Docker

First complete Steps 1-4 so the trained models exist in backend/artifacts/. Then, from the project root:

```
docker compose up --build
```

Open **http://localhost:8080**. To pass your API key to Docker, create a file named **.env** in the project root (next to docker-compose.yml) containing **FOOTBALL_DATA_ORG_KEY=your_key**. Stop with **Ctrl + C** or **docker compose down**.

7. Settings (optional)

All settings live in **backend/.env**. Change a value, save, and restart the backend. The most useful:

Setting	Default	Purpose
FOOTBALL_DATA_ORG_KEY	(empty)	Champions League fixtures, results and live scores
MP_SCHEDULER_ENABLED	true	Automatic daily ingest, live refresh and weekly retrain
MP_INGEST_HOUR_UTC	7	Hour (UTC) of the daily results/fixtures update
MP_LIVE_REFRESH_MINUTES	2	How often live scores are refreshed during matches
MP_PORT	8000	Port of the backend API
MP_CORS_ORIGINS	http://localhost:5173	Website addresses allowed to call the API
MP_RATE_LIMIT	60/minute	Requests allowed per visitor

The full list with explanations is in backend/.env.example.

8. Troubleshooting

Problem	Solution
"No prediction available" on matches	Models are not trained or loaded. Run Step 4, then restart the backend. Check http://127.0.0.1:8000/api/v1/health shows "model_loaded": true.
No Champions League matches	The API key is missing or wrong in backend/.env. Then run: uv run python -m matchpredict.ingest.europe refresh
Only a few domestic fixtures	The data source publishes only the next round (Tuesdays and Fridays). The daily update picks up new rounds automatically.
Website opens but shows no data	Make sure the backend terminal is running, and open http://localhost:5173 (not 127.0.0.1).
"Port 8000/5173 already in use"	Another copy is still running. Close the old terminal, or set MP_PORT=8001 in backend/.env and update the proxy in frontend/vite.config.ts.
"uv" or "npm" not recognized	The tool is not installed or not on PATH. Reinstall it and open a new terminal.
SSL / handshake errors while downloading	The app retries automatically and falls back to curl. Make sure curl is installed (it is built into Windows 10/11 and macOS).
Training is slow	It uses all CPU cores and takes about 20 minutes. Use --quick for a fast trial run.

9. For developers

Run the tests

```
# backend
cd backend
uv run pytest
uv run ruff check src tests

# frontend
cd frontend
npm test
npm run lint
npm run build
```

Project layout

```
backend/
  src/matchpredict/
    predictor/   ratings engine, features, models, training, prediction service
    ingest/      data sources (football-data.co.uk, football-data.org,
                 TheSportsDB, openfootball) and the scheduler
    db/          database models, queries, seed
    api/         FastAPI app and routes (/api/v1)
  tests/
  artifacts/     trained models (created by training, not in git)
frontend/
  src/           React app: pages, components, API client, utilities
data/raw/        Matches.csv and EloRatings.csv (not in git)
docker-compose.yml
```

Data sources and credits

Source	Used for
football-data.co.uk	Historical results, statistics and odds; daily results and fixtures
football-data.org	Champions League schedule, results and live scores (free API key)
openfootball	UEFA competition results 2011-2026 (CC0 licence)
TheSportsDB	Europa / Conference League fixtures (fallback)
ClubElo	Cross-country club ratings (EloRatings.csv)

Data belongs to its providers; check their terms before any commercial use. The project code is released under the MIT License (c) 2026 Shoaib Ghulam.